Chapter 4
Limits and Continuity

Note 4

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4.3 : Continuity ctd.

What more can be said about continuous functions?

One useful conceptualization of functions which are continuous on an interval is that their graphs have no holes or jumps (see **Theorem 4.3.5** below). Our proof of this fact is based on the following elementary observation.

Lemma : Suppose that $a < b$ and that $f : [a, b) \to \mathbb{R}$. If f is continuous at a point $x_0 \in [a, b)$ and $f(x_0) > 0$, then there exist a positive number ε and a point $x_1 \in [a, b)$ such that $x_1 > x_0$ and $f(x) > \varepsilon$ for all $x \in [x_0, x_1]$.

Strategy : The idea behind the proof is simple. If $f(x_0) > 0$, then $f(x) > \dfrac{f(x_0)}{2}$ for x near x_0. Here are the details.

Proof of the Lemma :

- Let $\varepsilon = \dfrac{f(x_0)}{2}$. Since $x_0 < b$, $\delta_0 := \dfrac{(b - x_0)}{2}$ is positive and $a \leq x < x_0 + \delta_0$ implies $x \in [a, b)$.
- Use **Definition 4.3.1** to choose $0 < \delta < \delta_0$ such that $x \in [a, b)$ and $|x - x_0| < \delta$ imply $|f(x) - f(x_0)| < \delta$.
- Fix $x_1 \in (x_0, x_0 + \delta)$ and suppose that $x \in [x_0, x_1]$.
- By the choice of ε and δ, it is clear that

$$\frac{-f(x_0)}{2} < f(x) - f(x_0) < \frac{f(x_0)}{2}.$$

- Solving the left-hand inequality for $f(x)$, we conclude that $f(x) > \dfrac{f(x_0)}{2} = \varepsilon$.

□

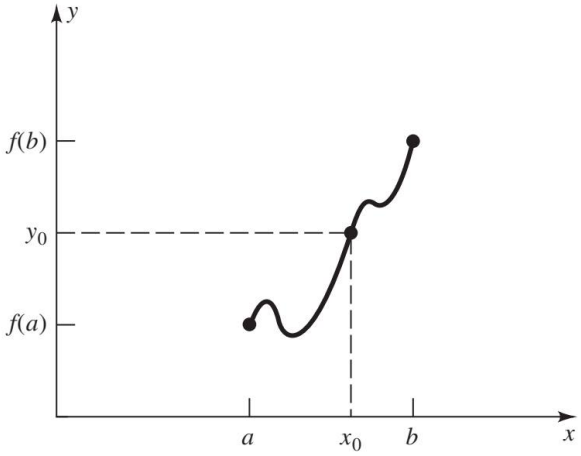
A real number y_0 is said to *lie between* two numbers c and d if and only if $c < y_0 < d$ or $d < y_0 < c$.

Theorem 4.3.5 : Intermediate Value Theorem (IVT)

Suppose that $a < b$ and that $f : [a, b] \to \mathbb{R}$ is continuous. If y_0 lies between $f(a)$ and $f(b)$, then there is an $x_0 \in (a, b)$ such that $f(x_0) = y_0$.

Proof of Theorem 4.3.5 :

- We may suppose that $f(a) < y_0 < f(b)$.
- Consider the set $E = \{x \in [a, b] \ : \ f(x) < y_0\}$



- Since $a \in E$ and $E \subseteq [a, b]$, E is a non-empty, bounded subset of $\mathbb{R}$.
- Hence, by the **Completeness Axiom**, $x_0 := \sup E$ is a finite number.
- It remains to prove that $x_0 \in (a, b)$ and $f(x_0) = y_0$.
- Choose by **Theorem 3.2.2** (Chapter 3-Note 2) a sequence $\{x_n\} \in E$ such that $x_n \to x_0$ as $n \to \infty$.
- Since $E \subseteq [a, b]$, it follows from **Theorem 3.2.5** (Chapter 3-Note 2) that $x_0 \in [a, b]$.
- Moreover, by the continuity of f and the definition of E, we have $f(x_0) = \lim_{n\to\infty} f(x_n) \leq y_0$.
- To show that $f(x_0) = y_0$, suppose to the contrary that $f(x_0) < y_0$.

- Then $y_0 - f(x)$ is a continuous function on the interval $[a, b)$ whose value at $x = x_0$ is positive.
- Hence, by the **Lemma**, we can choose an ε and an $x_1 > x_0$ such that $y_0 - f(x_1) > \varepsilon > 0$.
- In particular, $x_1 \in E$ and $x_1 > \sup E$, a contradiction.
- We have shown that $x_0 \in [a, b]$ and $y_0 = f(x_0)$.
- In view of our opening assumption, $f(a) < y_0 < f(b)$, it follows that x_0 cannot equal a or b.
- We conclude that $x_0 \in (a, b)$.

□

Remark :

- If f is continuous on $[a, b]$ and $f(a) \leq y_0 \leq f(b)$, then there is an $x_0 \in [a, b]$ such that $f(x_0) = y_0$.
- If f fails to be continuous at a point a, we say that f is *discontinuous* at a and call a *point of discontinuity* of f.
- How badly can a function behave near a point of discontinuity? The following examples can be interpreted as answers to this question.

Example 1 Prove that the function

$$f(x) = \begin{cases} \dfrac{|x|}{x} & : \ x \neq 0 \\ 1 & : \ x = 0 \end{cases}$$

is continuous on $(-\infty, 0)$ and $[0, \infty)$, discontinuous at 0, and that both $f(0^+)$ and $f(0^-)$ exist.

Proof .

- Since $f(x) = 1$ for $x \geq 0$, $f(0^+) = 1$ exists and $f(x) \to f(a)$ as $x \to a$ for any $a > 0$.
- In particular, f is continuous on $[0, \infty)$.
- Similarly, $f(0^-) = -1$ and f is continuous on $(-\infty, 0)$.
- Finally, since $f(0^+) \neq f(0^-)$, the limit of $f(x)$ as $x \to 0$ does not exist by **Theorem 4.2.1** (Chapter 4-Note 2). Therefore, f is not continuous at 0.

□

Example 2 Assuming that $\sin x$ is continuous on $\mathbb{R}$, prove that the function

$$f(x) = \begin{cases} \sin \dfrac{1}{x} & : \ x \neq 0 \\ 1 & : \ x = 0 \end{cases}$$

is continuous on $(-\infty, 0)$ and $(0, \infty)$, discontinuous at 0, and neither $f(0^+)$ nor $f(0^-)$ exists.

Proof.

- The function $\dfrac{1}{x}$ is continuous for $x \neq 0$.
- Hence, by **Theorem 4.3.3** (Chapter 4-Note 3), $f(x) = \sin \dfrac{1}{x}$ is continuous on $(-\infty, 0)$ and $(0, \infty)$.
- To prove that $f(0^+)$ does not exist, let $x_n = \dfrac{2}{((2n+1)\pi)}$, and observe that $\sin \dfrac{1}{x} = (-1)^n$, $n \in \mathbb{N}$.
- Since $x_n \downarrow 0$ but $(-1)^n$ does not converge, it follows from **Theorem 4.3.1 (The Sequential Characterization of Continuity)** that $f(0^+)$ does not exist.
- A similar argument proves that $f(0^-)$ does not exist.

□

In the next two examples, we introduce functions that played a significant role in the development of real analysis. Discontinuities are emphasized and it is not possible to graph either of them satisfactorily. The intuitive idea of drawing a curve in the plane to represent a function simply does not apply, and plotting a handful of points gives only a hint of their character. In the 19th century, these functions clearly demonstrated the need for a precise and rigorous treatment of the basic concepts of analysis.

Example 3 The *Dirichlet function* is defined on $\mathbb{R}$ by

$$f(x) := \begin{cases} 1 & : \ x \in \mathbb{Q} \\ 0 & : \ x \notin \mathbb{Q} \end{cases}$$

Prove that every point $x \in \mathbb{R}$ is a point of discontinuity of f. (Such functions are called *nowhere continuous*). This function was introduced in 1829 by P. G. L. Dirichlet.

Solution *:*

Method 1

- By the **Density of Rationals and Irrationals**, given any $a \in \mathbb{R}$ and $\delta > 0$ we can choose $x_1 \in \mathbb{Q}$ and $x_2 \in \mathbb{R}\backslash\mathbb{Q}$ such that $|x_i - a| < \delta$ for $i = 1, 2$.
- Since $f(x_1) = 1$ and $f(x_2) = 0$, f cannot be continuous at a, namely does not satisfy (5) (in **Definition 5.3.1** in Chapter 5-Note 3) for any $\varepsilon < 1$.

Method 2

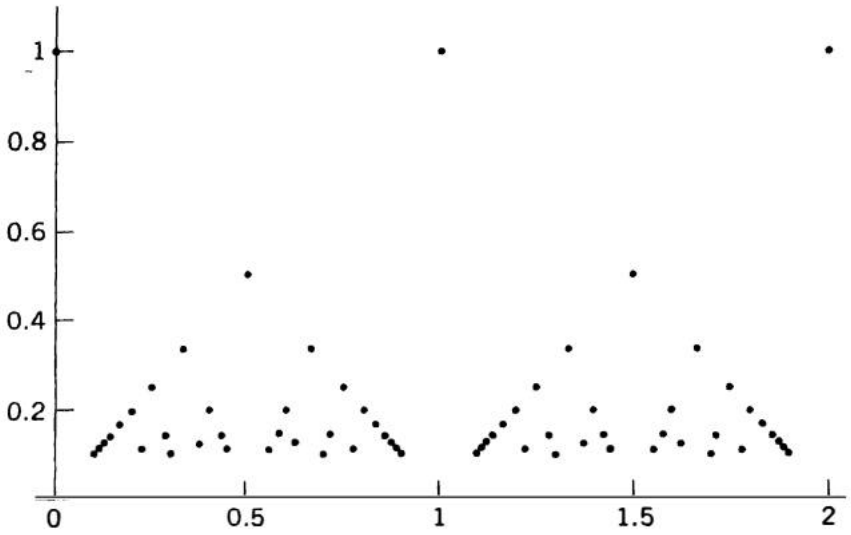
- Let c be any real number.
- If $c \in \mathbb{Q}$, let $\{x_n\}$ be a sequence of irrational numbers that converges to c. Since $f(x_n) = 0$ for all $n \in \mathbb{Z}^+$, we have $\lim\limits_{n \longrightarrow \infty} f(x_n) = 0$, while $f(c) = 1$. Therefore f is not continuous at the rational number c.
- If $c \in \mathbb{R}\backslash\mathbb{Q}$, let $\{y_n\}$ be a sequence of rational numbers that converge to c. Since $f(y_n) = 1$ for all $n \in \mathbb{Z}^+$, we have $\lim\limits_{n \longrightarrow \infty} f(y_n) = 1$, while $f(c) = 0$. Therefore f is not continuous at the irrational number c.
- Since every real number is either rational or irrational, we deduce that f is not continuous at any point in $\mathbb{R}$.

□

Example 4 Prove that the function

$$f(x) := \begin{cases} \dfrac{1}{q} & : \ x = \dfrac{p}{q} \in \mathbb{Q} \ \text{ (in reduced form)} \\ 0 & : \ x \notin \mathbb{Q} \end{cases}$$

is continuous at every irrational in the interval $(0, 1)$ but discontinuous at every rational in $(0, 1)$. This function was introduced in 1875 by K. J. Thomae.



Solution *:*

- Let a be a rational in $(0, 1)$ and suppose that f is continuous at a.
- If $\{x_n\}$ is a sequence of irrationals which converges to a, then $f(x_n) \to f(a)$; that is $f(a) = 0$. But $f(a) \neq 0$ by definition. Hence, f is discontinuous at every rational in $(0, 1)$.
- Let a be an irrational in $(0, 1)$.
- We must show that $f(x_n) \to f(a)$ for every sequence $\{x_n\} \in (0, 1)$ which satisfies $x_n \to a$ as $n \to \infty$.
- We may suppose that $\{x_n\} \in \mathbb{Q}$. For each $n \in \mathbb{Z}^+$, write $x_n = \dfrac{p_n}{q_n}$ in reduced form.
- Since $f(a) = 0$, it suffices to show that $q_n \to \infty$ as $n \to \infty$.

- Suppose to the contrary that there exist integers $n_1 < n_2 < \dots$ such that $|q_{n_k}| \leq M < \infty$ for $k \in \mathbb{Z}^+$.

- Since $x_{n_k} \in (0,1)$, it follows that the set

$$E := \left\{ x_{n_k} = \frac{p_{n_k}}{q_{n_k}} \ : \ k \in \mathbb{Z}^+ \right\}$$

contains only a finite number of points.

- Hence, the limit of any sequence in E must belong to E, a contradiction since a is such a limit and is irrational.

□

Remark : The composition of two functions $g \circ f$ can be nowhere continuous, even though f is discontinuous only on $\mathbb{Q}$ and g is discontinuous at only one point.

Proof of the Remark *:*

- Let f be the function given in **Example 4** and set

$$g(x) = \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases}$$

- Clearly,

$$(g \circ f)(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q}. \end{cases}$$

- Hence, $g \circ f$ is the Dirichlet function, nowhere continuous by **Example 3**.

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EXERCISES

(1). Decide which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

(a). If f is continuous on $[a, b]$ and $J := f([a, b])$, then j is a closed, bounded interval.

(b). If f and g are continuous on $[a, b]$, if $f(a) < g(a)$ and $f(b) > g(b)$, then there is a $c \in [a, b]$ such that $f(c) = g(c)$.

(c). Suppose that f and g are defined and finite valued on an open interval I which contains a, that f is continuous at a, and that $f(a) \neq 0$. Then g is continuous at a if and only if fg is continuous at a.

(d). Suppose that f and g are defined and finite valued on $\mathbb{R}$. If f and $g \circ f$ are continuous on $\mathbb{R}$, then g is continuous on $\mathbb{R}$.

(2). Use limit theorems to show that the following functions are continuous on $[0, 1]$.

(a). $f(x) = \dfrac{e^{x^2}\sqrt{\sin x}}{\cos x}$

(b). $f(x) = \begin{cases} \dfrac{x^2 + x - 2}{x - 1} & : \quad x \neq 1 \\ 3 & : \quad x = 1 \end{cases}$

(c). $f(x) = \begin{cases} e^{-\frac{1}{x}} & : \quad x \neq 0 \\ 0 & : \quad x = 0 \end{cases}$

(d). $f(x) = \begin{cases} \sqrt{x}\ \sin\dfrac{1}{x} & : \quad x \neq 0 \\ 0 & : \quad x = 0 \end{cases}$

(3). For each of the following, prove that there is at least one $x \in \mathbb{R}$ which satisfies the given equation.

(a). $e^x = x^3$ **(b).** $e^x = 2\cos x + 1$ **(c).** $2^x = 2 - 3x$

(4). Suppose that $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$ for each $x, y \in \mathbb{R}$.

(a). Show that $f(nx) = nf(x)$ for all $x \in \mathbb{R}$ and $n \in \mathbb{Z}$.

(b). Prove that $f(qx) = qf(x)$ for all $x \in \mathbb{R}$ and $q \in \mathbb{Q}$.

(c). Prove that f is continuous at 0 if and only if f is continuous on $\mathbb{R}$.

(d). Prove that if f is continuous at 0, then there is an $m \in \mathbb{R}$ such that $f(x) = mx$ for all $x \in \mathbb{R}$.

(5). If $f : [a, b] \to \mathbb{R}$ is continuous, prove that $\sup_{x\in[a,b]} |f(x)|$ is finite.

(6). If $f : [a, b] \to [a, b]$ is continuous, then f has a fixed point; that is, there is a $c \in [a, b]$ such that $f(c) = c$.

(7). If f is a real function which is continuous at $a \in \mathbb{R}$ and if $f(a) < M$ for some $M > 0$, prove that there is an open interval I containing a such that $f(x) < M$ for all $x \in I$.

(8). Suppose that $a \in \mathbb{R}$, that I is an open interval containing a, that $f, g : I \to \mathbb{R}$, and that f is continuous at a. Prove that g is continuous at a if and only if $f + g$ is continuous at a.

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